



IMDG - UN card

UN 3478

UN number 3478		Shipping name FUEL CELL CARTRIDGES / FUEL CELL CARTRIDGES CONTAINED IN EQUIPMENT / FUEL CELL CARTRIDGES PACKED WITH EQUIPMENT containing liquefied flammable gas BRÆNDSSELCELEPATRONER / BRÆNDSSELCELEPATRONER INDEHOLDT I Udstyr / BRÆNDSSELCELEPATRONER PAKKET MED Udstyr				
Class 2.1	Packing group —	Hazard labels 2.1	Marine pollutant Maybe	EmS F-D, S-U	ADR transport category: 2 ADR tunnel: (D)	
Hazard labels 		Marking UN 3478 FUEL CELL CARTRIDGES / FUEL CELL CARTRIDGES CONTAINED IN EQUIPMENT / FUEL CELL CARTRIDGES PACKED WITH EQUIPMENT containing liquefied flammable gas				
Packaging		Can be packed in approved packaging according to packing instruction P004.				
Mixed packaging		The substance may be packaged with other substances in the same outer packaging if: - it is permitted to load the substances together, and - the substances do not react dangerously with one another.				
Remarks		This entry applies to fuel cell cartridges, including when contained in equipment or packed with equipment. Fuel cell cartridges installed in or integral to a fuel cell system are regarded as contained in equipment. "Fuel cell cartridge" means an article that stores fuel for discharge into the fuel cell through a valve(s) that controls the discharge of fuel into the fuel cell. Fuel cell cartridges, including when contained in equipment, shall be designed and constructed to prevent fuel leakage under normal conditions of transport. Fuel cell cartridge design types using liquids as fuels shall pass an internal pressure test at a pressure of 100 kPa (gauge) without leakage. Except for fuel cell cartridges containing hydrogen in metal hydride, which shall be in compliance with special provision 339, each fuel cell cartridge design type shall be shown to pass a 1.2 m drop test onto an unyielding surface, in the orientation most likely to result in failure of the containment system, with no loss of contents. When lithium metal or lithium ion batteries are contained in the fuel cell system, the consignment shall be consigned under this entry and under the appropriate entries for UN 3091 LITHIUM METAL BATTERIES CONTAINED IN EQUIPMENT or UN 3481 LITHIUM ION BATTERIES CONTAINED IN EQUIPMENT (SP 328). Each fuel cell cartridge transported under this entry and designed to contain a liquefied flammable gas shall: 1. be capable of withstanding, without leakage or bursting, a pressure of at least two times the equilibrium pressure of the contents at 55°C; 2. not contain more than 200 mL liquefied flammable gas, the vapour pressure of which shall not exceed 1 000 kPa at 55°C; and 3. pass the hot water bath test prescribed in 6.2.4.1 of chapter 6.2 (SP 338).				
Properties and observations		Fuel cell cartridges containing butane or other flammable liquefied gas may also be shipped in, or packed with, equipment.				
Tank type:	Not allowed	Transport in bulk			Orange panel	
Tank provisions:	—	Not allowed			3478	
Stowage and segregation						
Stowage		There are no special requirements for stowage.				
Segregation		Check the segregation table for sea transport. There are no special requirements for segregation.				
Placarding of the transport unit		Containers, semi-trailers and tank containers must be marked with enlarged hazard labels (placards) on all four sides. Railway wagons must at least be marked with placards on both sides. All other transport units, e.g. trucks etc., must at least be placarded on both sides and at the back. The placards must have a size of at least 250 x 250 mm, and be of such quality that the labeling is still visible after 3 months at sea.				
Foodstuff		The substance is not required to be kept separate from food and feed.				
Limited quantities (LQ) <i>Combination packaging</i> Inner: 120 mL Outer: 30 kg <i>Trays</i> Inner: 120 mL Outer: 20 kg		Parcel label: 		Excepted quantities (EQ) <i>Combination packaging</i> Inner: Not allowed Outer:		